

HAT-P-20b

R-1334

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_220104 · created 2026-07-31T20:56:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459583.6709 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1185 +/- 0.0057
Transit depth (Rp/Rs)^2	1.4 +/- 0.13 [%]
Orbital Inclination (inc)	89.97 +/- 0.5
Airmass coefficient 1 (a1)	1.1219 +/- 0.0077
Airmass coefficient 2 (a2)	-0.0915 +/- 0.0012
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	5.24
Optimal Annulus	12.51
Transit Duration (day)	0.09216 +/- 0.00068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1185 $\pm$ 0.0057 vs 0.1284 (deviation 1.6 σ)
Duration fit vs published	0.09216 d vs 0.0775 d (deviation 19.82 σ)
Mid-time O-C	-6.0 min
Depth SNR	19.1
Residual scatter	0.67 %

Light curve

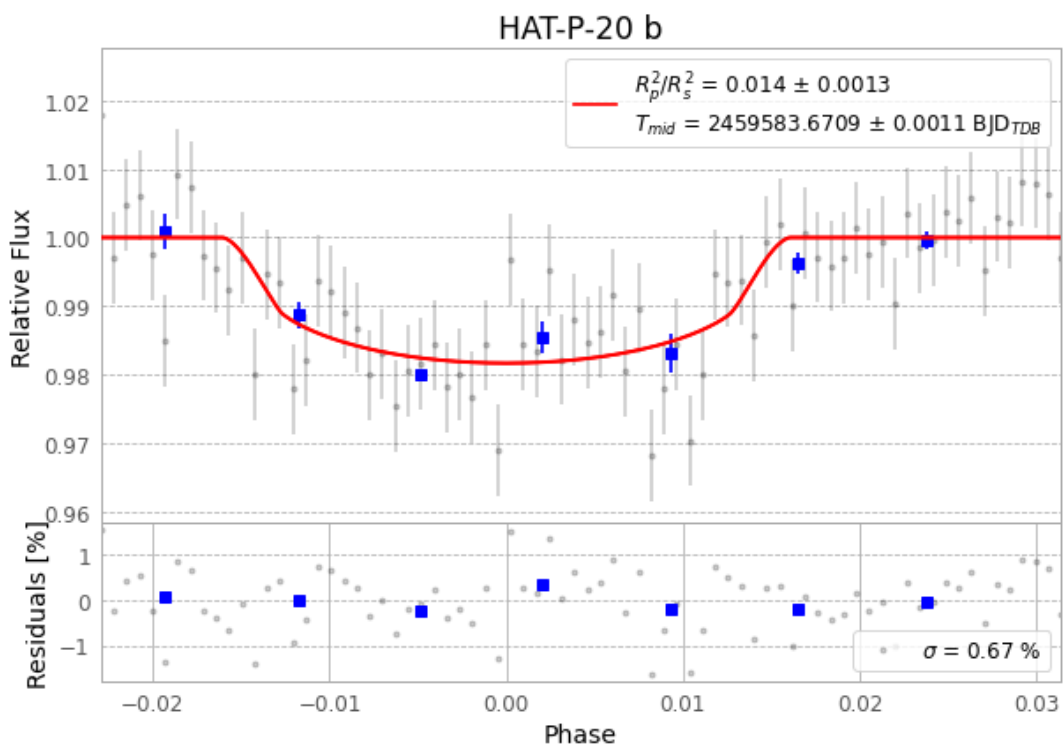


Figure 1 — FinalLightCurve_HAT-P-20 b_2022-01-03.png (SHA-256 691e683e10508b6d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cc540b3bc9727f12...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), HATP-20220104022112.FITS → HATP-20220104060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-220104.log (24cf4016f150...), FinalLightCurve_HAT-P-20 b_2022-01-03.png (691e683e1050...), FinalLightCurve_HAT-P-20 b_2022-01-03.csv (83623bda33c1...)

Compute resources

Wall time 210.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:56Z.